

FICHTELBERG, Germany

WMO: 105780

Lat: 50.4283N

Lon: 12.9536E

Elev: 3985

StdP: 12.70

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.2	5.8	-12.4	3.3	20.0	-4.9	4.9	18.2	56.2	32.1	51.6	30.7	17.4	90	1.049

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.3	70.8	58.8	67.6	57.2	64.7	55.7	60.3	67.9	58.4	65.0	56.9	62.8	12.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.4	81.7	62.7	55.8	76.9	60.6	54.4	73.1	59.3	28.5	68.2	27.2	65.1	26.1	62.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
48.8	44.0	39.9	DB	-0.2	77.8	9.1	3.5	-6.8	80.3	-12.1	82.4	-17.2	84.4	-23.8	86.9	
			WB	-0.6	64.2	8.9	1.6	-7.0	65.4	-12.3	66.3	-17.3	67.2	-23.8	68.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	24.3	23.0	28.6	34.6	44.2	49.2	53.8	54.1	47.3	40.1	31.2	27.4	(6)
(7)		DBStd	13.39	8.95	9.76	7.88	7.68	7.49	7.29	6.99	6.96	6.70	7.70	7.95	7.31	(7)
(8)		HDD50	4779	796	755	662	464	206	103	41	39	131	319	563	700	(8)
(9)		HDD65	9766	1261	1175	1127	912	644	474	350	343	530	772	1013	1165	(9)
(10)		CDD50	496	0	0	0	1	27	80	160	166	51	11	0	0	(10)
(11)		CDD65	10	0	0	0	0	0	1	4	5	0	0	0	0	(11)
(12)		CDH74	30	0	0	0	0	0	1	17	12	0	0	0	0	(12)
(13)		CDH80	3	0	0	0	0	0	0	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	20.3	26.1	22.6	21.2	17.9	16.9	16.5	15.9	16.1	19.5	22.8	23.6	24.8	(14)
(15)	Precipitation	PrecAvg	45.2	3.7	3.1	3.5	3.0	3.8	4.3	4.7	4.3	3.7	3.1	3.7	4.3	(15)
(16)		PrecMax	66.3	9.6	8.1	9.5	7.3	7.8	9.5	9.6	11.8	9.3	6.7	8.6	10.9	(16)
(17)		PrecMin	30.3	0.5	0.5	0.9	0.3	1.2	2.0	0.9	0.8	0.6	0.2	0.0	0.3	(17)
(18)		PrecStd	8.5	1.9	1.9	1.9	1.4	1.7	1.5	1.9	2.1	1.9	1.6	1.9	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.1	47.3	52.2	56.9	64.6	70.9	76.3	75.5	67.2	61.7	53.5	47.0	(19)
(20)			MCWB	35.3	36.9	41.7	45.3	52.0	58.6	61.7	61.3	56.3	50.7	41.2	36.5	(20)
(21)		2%	DB	39.7	43.1	46.1	53.2	61.0	66.6	71.0	71.0	63.4	57.2	49.3	43.0	(21)
(22)			MCWB	32.1	36.1	40.2	44.4	51.0	56.6	58.6	59.4	55.0	47.5	39.9	35.9	(22)
(23)		5%	DB	37.0	39.7	42.1	49.6	57.8	63.1	67.8	67.5	60.2	53.7	45.7	40.1	(23)
(24)			MCWB	32.3	34.8	37.3	42.3	49.4	54.8	57.3	57.6	53.3	46.8	39.8	34.5	(24)
(25)		10%	DB	34.7	36.3	38.6	46.0	55.0	60.0	64.4	64.3	57.1	50.2	42.3	37.7	(25)
(26)			MCWB	31.5	32.0	35.9	40.8	47.9	52.9	55.7	55.9	51.7	44.9	38.2	34.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.7	40.6	45.2	48.1	54.8	60.5	63.0	63.0	58.3	53.5	44.9	42.0	(27)
(28)			MCDWB	40.8	44.0	49.8	54.1	61.2	67.4	72.3	72.9	63.7	58.8	46.9	42.7	(28)
(29)		2%	WB	36.0	38.0	40.8	45.6	52.3	58.2	60.4	60.4	56.3	50.7	42.8	39.0	(29)
(30)			MCDWB	37.0	40.3	44.1	50.6	57.8	63.9	68.0	68.6	61.2	54.6	45.3	40.5	(30)
(31)		5%	WB	34.2	35.0	38.2	43.5	50.5	56.1	58.4	58.4	54.6	48.4	41.1	36.8	(31)
(32)			MCDWB	35.1	38.5	41.0	48.1	56.3	61.3	64.9	65.5	58.8	50.9	44.6	38.1	(32)
(33)		10%	WB	32.3	32.5	36.2	41.1	48.7	54.1	56.7	56.8	52.7	46.8	39.0	34.9	(33)
(34)			MCDWB	33.6	35.1	38.5	45.3	54.2	59.0	62.7	62.6	56.4	49.0	41.8	36.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	8.2	8.1	9.9	11.6	10.9	12.0	11.3	9.9	8.9	7.6	7.5	(35)
(36)			MCDBR	10.8	10.8	12.2	14.3	15.2	14.4	16.7	15.4	13.6	13.1	11.2	10.5	(36)
(37)			MCWBR	8.2	7.7	8.4	8.7	8.6	8.7	9.1	8.9	8.4	8.3	8.1	7.2	(37)
(38)		5% WB	MCDBR	9.3	9.6	11.3	13.4	13.9	13.4	14.5	13.7	12.3	11.4	9.9	8.5	(38)
(39)			MCWBR	8.8	7.8	8.6	9.0	8.8	8.7	9.2	8.9	8.4	8.5	8.4	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.277	0.311	0.343	0.346	0.346	0.357	0.351	0.325	0.305	0.282	0.256	(40)	
(41)		taud	2.455	2.440	2.426	2.386	2.394	2.403	2.368	2.404	2.467	2.523	2.532	2.476	(41)	
(42)		Ebn at Noon	256	278	285	286	289	289	283	280	277	264	245	240	(42)	
(43)		Edh at Noon	18	24	30	34	36	36	36	33	28	22	17	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	258	512	817	1279	1531	1662	1582	1377	985	586	280	200	(44)	
(45)		RadStd	49	77	117	167	186	143	172	134	138	75	35	24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	+0.72	N/A	N/A	N/A	+0.58	+0.63	N/A	N/A	+94	+29

Nomenclature: See separate page